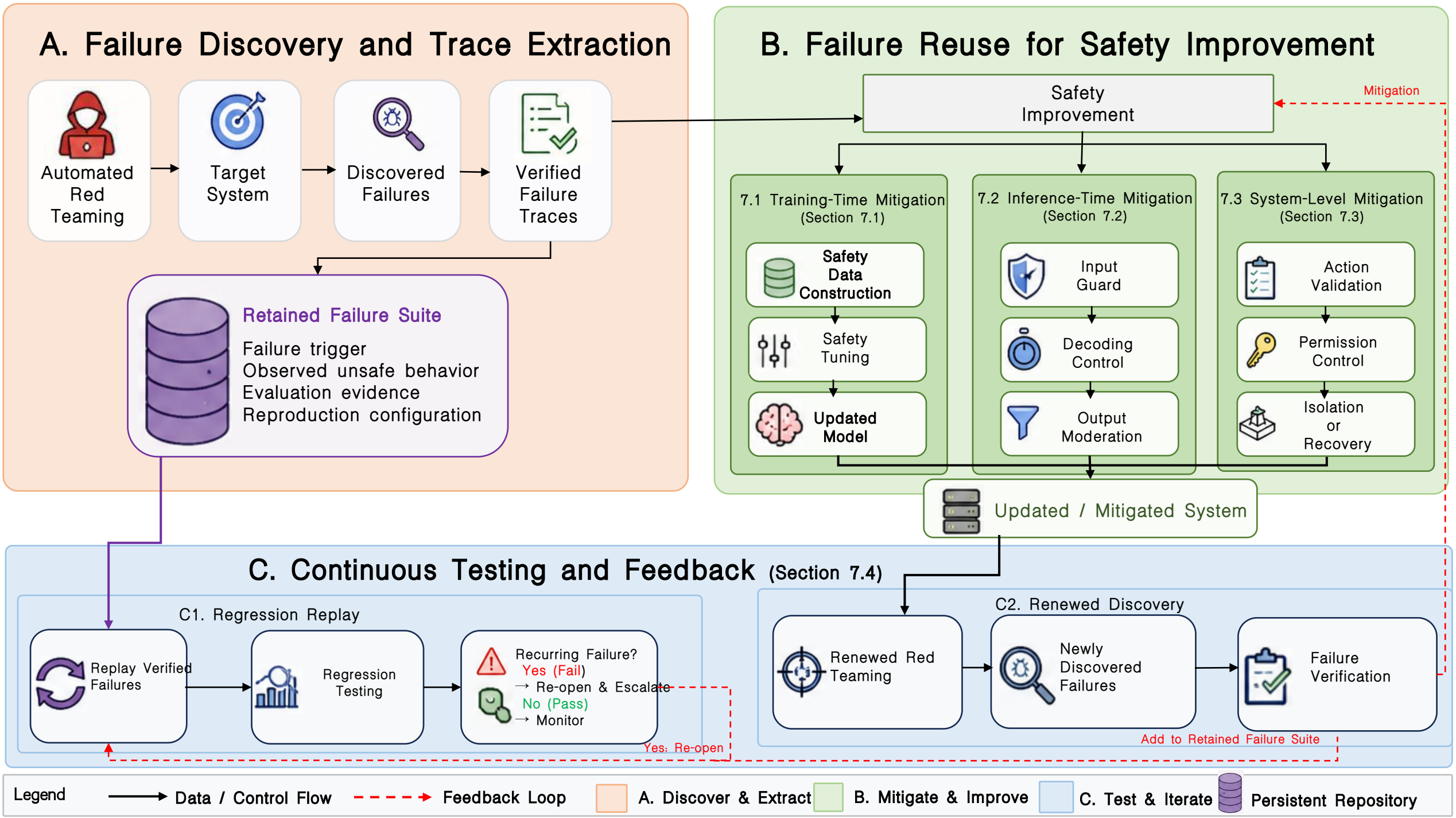

A. Failure Discovery and Trace Extraction
Automated Red Teaming
Target System
Discovered Failures
Verified Failure Traces
Retained Failure Suite
Failure trigger
Observed unsafe behavior
Evaluation evidence
Reproduction configuration
B. Failure Reuse for Safety Improvement
Safety Improvement
Mitigation
7.1 Training-Time Mitigation (Section 7.1)
Safety Data Construction
Safety Tuning
Updated Model
7.2 Inference-Time Mitigation (Section 7.2)
Input Guard
Decoding Control
Output Moderation
7.3 System-Level Mitigation (Section 7.3)
Action Validation
Permission Control
Isolation or Recovery
Updated / Mitigated System
C. Continuous Testing and Feedback (Section 7.4)
C1. Regression Replay
Replay Verified Failures
Regression Testing
Recurring Failure?
Yes (Fail)
→ Re-open & Escalate
No (Pass)
→ Monitor
Yes: Re-open
C2. Renewed Discovery
Renewed Red Teaming
Newly Discovered Failures
Failure Verification
Add to Retained Failure Suite
Legend
Data / Control Flow
Feedback Loop
A. Discover & Extract
B. Mitigate & Improve
C. Test & Iterate
Persistent Repository